

Exercise 8.10

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Derive an eq. of the cylinder from definition with given direction & elements parallel to the given vector (Problems 1-3).

Q1 $x^2 + y^2 = 9$, $\vec{n} = [1, -2, 1]$

Sol. The eq. of line L through (x_1, y_1, z_1) & parallel to $\vec{n}$ is

$$\frac{x-x_1}{1} = \frac{y-y_1}{-2} = \frac{z-z_1}{1} = t \text{ (say)}$$

$$\Rightarrow x = x_1 + t \quad \text{--- (i)}$$

$$y = y_1 - 2t \quad \text{--- (ii)}$$

$$z = t \quad \text{--- (iii)}$$

From (iii) $t = z$

Put in (i) & (ii)

$$\left. \begin{aligned} x &= x_1 + z \\ y &= y_1 - 2z \end{aligned} \right\}$$

$$\Rightarrow \left. \begin{aligned} x_1 &= x - z \\ y_1 &= y + 2z \end{aligned} \right\}$$

Since (x_1, y_1, z_1) lies on $x^2 + y^2 = 9$

$$\text{So } x_1^2 + y_1^2 = 9$$

$$(x-z)^2 + (y+2z)^2 = 9$$

$$x^2 - 2xz + z^2 + y^2 + 4yz + 4z^2 = 9$$

$$\boxed{x^2 + y^2 + 5z^2 - 2xz + 4yz = 9} \text{ is rep. eq. of cylinder}^{172}$$

Q2 $x+z=4$, $\vec{n} = [0, 2, -1]$

Sol. The eq. of line L through $(x_1, 0, z_1)$ & || to $\vec{n}$ is

$$\frac{x-x_1}{0} = \frac{y-0}{2} = \frac{z-z_1}{-1} = t \text{ (say)}$$

$$\Rightarrow x = x_1 \text{ ————— (i)}$$

$$y = 2t \text{ ————— (ii)}$$

$$z = z_1 - t \text{ ————— (iii)}$$

From (ii) $t = \frac{y}{2}$

Put in (i) & (iii)

$$\left. \begin{aligned} x &= x_1 \\ z &= z_1 - \frac{y}{2} \end{aligned} \right\}$$

$$\text{or } \left. \begin{aligned} x_1 &= x \\ z_1 &= z + \frac{y}{2} \end{aligned} \right\}$$

Since $(x_1, 0, z_1)$ lies on $x+z=4$

$$\text{So } x_1 + z_1 = 4$$

using values of x_1 & z_1

$$x + z + \frac{y}{2} = 4$$

$$2x + z + y = 8$$

$$\boxed{2x + y + z = 8} \text{ is rep. eq. of cylinder}$$

Q3. $\frac{z^2}{4} + \frac{y^2}{9} = 1$; $\vec{n} = [1, 1, 1]$

Sol. The eq. of line through $(0, y_1, z_1)$ & || to $\vec{n}$ is

$$\frac{x-0}{1} = \frac{y-y_1}{1} = \frac{z-z_1}{1} = t \text{ (say)}$$

$$\Rightarrow x = t \quad \text{--- (i)}$$

$$y = y_1 + t \quad \text{--- (ii)}$$

$$z = z_1 + t \quad \text{--- (iii)}$$

From (i) $t = x$

Put in (ii) & (iii)

$$\left. \begin{aligned} y &= y_1 + x \\ z &= z_1 + x \end{aligned} \right\}$$

$$\text{or } \left. \begin{aligned} y_1 &= y - x \\ z_1 &= z - x \end{aligned} \right\}$$

Since $(0, y_1, z_1)$ lies on $\frac{z^2}{4} + \frac{y^2}{9} = 1$

$$\text{So } \frac{z_1^2}{4} + \frac{y_1^2}{9} = 1$$

using values of y_1 & z_1

$$\frac{(z-x)^2}{4} + \frac{(y-x)^2}{9} = 1$$

$$9(z-x)^2 + 4(y-x)^2 = 36$$

$$9(z^2 + x^2 - 2zx) + 4(y^2 - 2xy + x^2) = 36$$

$$9z^2 + 9x^2 - 18xz + 4y^2 - 8xy + 4x^2 = 36$$

$$\boxed{13x^2 + 4y^2 + 9z^2 - 8xy - 18xz - 36 = 0} \text{ is req. eq.}$$

Q12 Write an eq. of right circular cylinder¹⁷⁴ with radius 2 & centre at $(3, 0, 5)$.

Sol:-

The eq. of circle with centre at $(3, 0, 5)$ & radius 2 is

$$(x-3)^2 + (z-5)^2 = (2)^2$$

or $(x-3)^2 + (z-5)^2 = 4$ is eq. of direction

which is also the eq. of right circular cylinder.

Q13 Write an eq. of the right elliptic cylinder whose direction is in the yz -plane with foci $(0, \pm 3, 0)$ & major axis 8.

Sol:- Here $2a = 8$

$$\Rightarrow a = 4$$

$$\& c = 3$$

For an ellipse

$$b^2 = a^2 - c^2$$

$$= (4)^2 - (3)^2$$

$$= 16 - 9$$

$$b^2 = 7$$

$$b = \sqrt{7}$$

The eq. of dTx which is an ellipse is

$$\frac{x^2}{16} + \frac{y^2}{7} = 1$$

$$\text{or } 7x^2 + 16y^2 = 112$$

$$\boxed{7x^2 + 16y^2 - 112 = 0}$$

which is also eq. of right elliptic cylinder.

Find an Eqn. of the Cone whose directrix & vertex are given (Problems 14-16):

Q14 Directrix : $y^2 + z^2 = 1$, $x = 2$ vertex $A(0,0,0)$

Sol. let $P(x,y,z)$ be a pt. on the req. Cone
& let line AP meets directrix at pt. $Q(x_1, y_1, z_1)$

Since pt. Q lies on directrix

$$\therefore y_1^2 + z_1^2 = 1, \quad x_1 = 2 \quad \text{--- (1)}$$

Then eq. of line AQ is

$$\frac{x-0}{x_1-0} = \frac{y-0}{y_1-0} = \frac{z-0}{z_1-0}$$

$$\frac{x}{x_1} = \frac{y}{y_1} = \frac{z}{z_1}$$

From (1) Put $x_1 = 2$

$$\frac{x}{2} = \frac{y}{y_1} \quad \therefore \frac{x}{2} = \frac{z}{z_1}$$

$$\text{or } \frac{y_1}{y} = \frac{2}{x}, \quad \frac{z_1}{z} = \frac{2}{x}$$

$$\text{or } y_1 = \frac{yz}{x}, \quad z_1 = \frac{zx}{x}$$

Put values in ①

$$\frac{4y^2}{x^2} + \frac{4z^2}{x^2} = 1$$

$$4y^2 + 4z^2 = x^2$$

$$\boxed{4(y^2 + z^2) = x^2} \text{ is req. eq. of Cone.}$$

Q15 Direction $4x^2 + (y-z)^2 = 4, z = 3$, Vertex $A(0,0,0)$

Soln — Let $P(x, y, z)$ be any pt. on req. Cone & let line AP meets direction at pt. $Q(x_1, y_1, z_1)$

Since $Q(x_1, y_1, z_1)$ lies on direction

$$\text{So } 4x_1^2 + (y_1 - z_1)^2 = 4, \quad z_1 = 3 \quad \text{--- ①}$$

The eq. of line AQ is

$$\frac{x-0}{x_1-0} = \frac{y-0}{y_1-0} = \frac{z-0}{z_1-0}$$

$$\therefore \frac{x}{x_1} = \frac{y}{y_1} = \frac{z}{z_1}$$

Put $z_1 = 3$

$$\frac{x}{x_1} = \frac{z}{3}, \quad \frac{y}{y_1} = \frac{z}{3}$$

$$\frac{x_1}{x} = \frac{3}{z}, \quad \frac{y_1}{y} = \frac{3}{z}$$

$$\Rightarrow x_1 = \frac{3x}{z}, \quad y_1 = \frac{3y}{z}$$

Put in ①

$$4\left(\frac{3x}{z}\right)^2 + \left(\frac{3y}{z} - 2\right)^2 = 4$$

$$4\left(\frac{9x^2}{z^2}\right) + \left(\frac{3y-2z}{z}\right)^2 = 4$$

$$\frac{36x^2}{z^2} + \frac{9y^2 - 12yz + 4z^2}{z^2} = 4$$

$$36x^2 + 9y^2 - 12yz + 4z^2 = 4z^2$$

$$36x^2 + 9y^2 - 12yz = 0$$

$$\boxed{12x^2 + 3y^2 - 4yz = 0} \text{ is req. eq. of Cone.}$$

Q16 DTX: $x^2 + 4y^2 - 2x + 8y - 4 = 0$; $z = 3$

Vertex $A = (-1, 2, 1)$

Sol. let $P(x, y, z)$ be any pt. on req. Cone & let line AP meets dTX at pt. $Q(x_1, y_1, z_1)$

Since pt. Q lies on dTX

$$\therefore x_1^2 + 4y_1^2 - 2x_1 + 8y_1 - 4 = 0; \quad z_1 = 3 \quad \text{--- ①}$$

The eq. of line AQ is

$$\frac{x+1}{x_1+1} = \frac{y-2}{y_1-2} = \frac{z-1}{z_1-1}$$

Put $z_1 = 3$

$$\frac{x+1}{x_1+1} = \frac{y-2}{y_1-2} = \frac{z-1}{z-1}$$

$$\frac{x+1}{x_1+1} = \frac{y-2}{y_1-2} = \frac{z-1}{2}$$

$$\frac{x+1}{x_1+1} = \frac{z-1}{2}, \quad \frac{y-2}{y_1-2} = \frac{z-1}{2}$$

$$\Rightarrow \frac{x+1}{x_1+1} = \frac{2}{z-1}, \quad \frac{y-2}{y_1-2} = \frac{2}{z-1}$$

$$x_1+1 = \frac{2x+2}{z-1}, \quad y_1-2 = \frac{2y-4}{z-1}$$

$$x_1 = \frac{2x+2}{z-1} - 1, \quad y_1 = \frac{2y-4}{z-1} + 2$$

$$x_1 = \frac{2x+2-z+1}{z-1}, \quad y_1 = \frac{2y-4+2z-2}{z-1}$$

$$x_1 = \frac{2x-z+3}{z-1}, \quad y_1 = \frac{2y+2z-6}{z-1}$$

Put in ①

$$\frac{(2x-z+3)^2}{(z-1)^2} + 4 \frac{(2y+2z-6)^2}{(z-1)^2} - 2 \frac{(2x-z+3)}{z-1} + 8 \frac{(2y+2z-6)}{(z-1)} - 4 = 0$$

$$(2x-z+3)^2 + 4(2y+2z-6)^2 - 2(2x-z+3)(z-1) + 16(y+z-3)(z-1) - 4(z-1)^2 = 0$$

$$4x^2 + z^2 + 9 - 4xz - (z + 12x + 16(y^2 + z^2 + 9 + 2yz - 6z - 6y))$$

$$- 2(2xz - z^2 + 3z - 2x + z - 3) + 16(yz + z^2 - 3z - y - z + 3) - 4(z^2 - 2z + 1) = 0$$

$$\begin{aligned}
 &4x^2 + z^2 + 9 - 4xz - 6z + 12x + 16y^2 + 16z^2 + 144 + 32yz \\
 &- 96z - 96y - 4xz + 2z^2 - 6z + 4x - 2z + 6 + 16yz + 16z^2 \\
 &- 48z - 16y - 16z + 48 - 4z^2 + 8z - 4 = 0
 \end{aligned}$$

$$4x^2 + 16y^2 + 31z^2 - 8xz + 48yz + 16x - 112y - 166z + 203 = 0$$

is req. eq. of Cone.

Q17 Show that the eq. of the Cone with vertex at $(3, 1, 2)$ & whose dir. $2x^2 + 3y^2 = 1, z = 0$

is $2x^2 + 3y^2 + 5z^2 - 3yz - 6zx + z - 1 = 0$

Sol. Let $P(x, y, z)$ be any pt. on req. Cone & let AP meets director at pt. $Q(x_1, y_1, z_1)$

Since $Q(x_1, y_1, z_1)$ lies on director

So $2x_1^2 + 3y_1^2 = 1, z_1 = 0$ ——— ①

The eq. of line AQ is

$$\frac{x-3}{x_1-3} = \frac{y-1}{y_1-1} = \frac{z-2}{z_1-2}$$

Put $z_1 = 0$

$$\frac{x-3}{x_1-3} = \frac{z-2}{-2}, \quad \frac{y-1}{y_1-1} = \frac{z-2}{-2}$$

$$\Rightarrow \frac{x_1-3}{x-3} = -\frac{2}{z-2}, \quad \frac{y_1-1}{y-1} = -\frac{2}{z-2}$$

$$x_1-3 = \frac{-2(x-3)}{z-2}, \quad y_1-1 = \frac{-2(y-1)}{z-2}$$

$$x_1 = \frac{6-2x}{z-2} + 3, \quad y_1 = \frac{2-2y}{z-2} + 1$$

$$x_1 = \frac{6-2x+3z-6}{z-2}, \quad y_1 = \frac{2-2y+z-2}{z-2}$$

$$x_1 = \frac{3z-2x}{z-2}, \quad y_1 = \frac{z-2y}{z-2}$$

Using these values in ①

$$2 \left(\frac{3z-2x}{z-2} \right)^2 + 3 \left(\frac{z-2y}{z-2} \right)^2 = 1$$

$$2 \frac{(3z-2x)^2}{(z-2)^2} + 3 \frac{(z-2y)^2}{(z-2)^2} = 1$$

$$2(3z-2x)^2 + 3(z-2y)^2 = (z-2)^2$$

$$2(9z^2 - 12zx + 4x^2) + 3(z^2 - 4yz + 4y^2) = z^2 - 4z + 4$$

$$18z^2 - 24zx + 8x^2 + 3z^2 - 12yz + 12y^2 - z^2 + 4z - 4 = 0$$

$$8x^2 + 12y^2 + 20z^2 - 24zx - 12yz + 4z - 4 = 0$$

$$\boxed{2x^2 + 3y^2 + 5z^2 - 6xz - 3yz + z - 1 = 0}$$

is req. eq.

Q18 Show that an eq. of the cylinder whose generators intersect the curve

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0, \quad z = 0$$

and are parallel to $\frac{x}{l} = \frac{y}{m} = \frac{z}{n}$ is

$$a(nx - lz)^2 + 2h(nx - lz)(ny - mz) + b(ny - mz)^2 + 2gn(nx - lz) + 2nf(ny - mz) + n^2c = 0$$

Sol. Let $P(x_1, y_1, z_1)$ be any pt. on the req. cylinder. The eq. of line through P & || to $\frac{x}{l} = \frac{y}{m} = \frac{z}{n}$ is

$$\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n} = t \quad \text{--- (1)}$$

$$\Rightarrow \left. \begin{aligned} x &= x_1 + lt \\ y &= y_1 + mt \\ z &= z_1 + nt \end{aligned} \right\}$$

Any pt. on line (1) is

$$(x_1 + lt, y_1 + mt, z_1 + nt)$$

If this pt. lies on direction

then

$$a(x_1 + lt)^2 + 2h(x_1 + lt)(y_1 + mt) + b(y_1 + mt)^2 + 2g(x_1 + lt) + 2f(y_1 + mt) + c = 0$$

$$\text{and } z_1 + nt = 0$$

$$\Rightarrow t = -\frac{z_1}{n}$$

Put in above eqⁿ

$$a\left(x_1 - \frac{lz_1}{n}\right)^2 + 2h\left(x_1 - \frac{lz_1}{n}\right)\left(y_1 - \frac{mz_1}{n}\right) + b\left(y_1 - \frac{mz_1}{n}\right)^2 \\ + 2g\left(x_1 - \frac{lz_1}{n}\right) + 2f\left(y_1 - \frac{mz_1}{n}\right) + C = 0$$

$$a\left(\frac{nx_1 - lz_1}{n}\right)^2 + 2h\left(\frac{nx_1 - lz_1}{n}\right)\left(\frac{ny_1 - mz_1}{n}\right) + b\left(\frac{ny_1 - mz_1}{n}\right)^2 \\ + 2g\left(\frac{nx_1 - lz_1}{n}\right) + 2f\left(\frac{ny_1 - mz_1}{n}\right) + C = 0$$

$$a(nx_1 - lz_1)^2 + 2h(nx_1 - lz_1)(ny_1 - mz_1) + b(ny_1 - mz_1)^2 \\ + 2gn(nx_1 - lz_1) + 2nf(ny_1 - mz_1) + n^2C = 0$$

Now the locus of (x_1, y_1, z_1) is

$$a(nx - lz)^2 + 2h(nx - lz)(ny - mz) + b(ny - mz)^2 \\ + 2gn(nx - lz) + 2nf(ny - mz) + n^2C = 0$$

which is eq. of req. Cylinder.

Q19 Show that an eq. of the cylinder whose generators are parallel to z -axis & which passes through the Curve $x^2 + y^2 + z^2 = 1$, $x + y + z = 1$ is $x^2 + y^2 + xy - x - y = 0$

Soln Let $P(x_1, y_1, z_1)$ be any pt. on req. Cylinder

Now eq. of line through (x_1, y_1, z_1) & $\parallel$ to z -axis is

$$\frac{x - x_1}{0} = \frac{y - y_1}{0} = \frac{z - z_1}{1} = t \quad \text{--- ①}$$

$$\Rightarrow \left. \begin{aligned} x &= x_1 \\ y &= y_1 \\ z &= z_1 + t \end{aligned} \right\}$$

Any pt. on line ① is $(x_1, y_1, z_1 + t)$
 If this pt. lies on director

$$x^2 + y^2 + z^2 = 1, \quad x + y + z = 1, \quad \text{so}$$

$$x_1^2 + y_1^2 + (z_1 + t)^2 = 1 \quad \text{--- ②}$$

$$x_1 + y_1 + z_1 + t = 1 \quad \text{--- ③}$$

From ③

$$t = 1 - x_1 - y_1 - z_1$$

Put in ②

$$x_1^2 + y_1^2 + (z_1 + 1 - x_1 - y_1 - z_1)^2 = 1$$

$$x_1^2 + y_1^2 + 1 + x_1^2 + y_1^2 - 2x_1 + 2x_1y_1 - 2y_1 = 1$$

$$2x_1^2 + 2y_1^2 + 2x_1y_1 - 2x_1 - 2y_1 = 0$$

$$x_1^2 + y_1^2 + x_1y_1 - x_1 - y_1 = 0$$

Now the locus of (x_1, y_1, z_1) is

$$x^2 + y^2 + xy - x - y = 0$$

which is eq. of required cylinder.

Q20 Show that an eq. of the right circular cylinder of radius 3 & whose axis is the line $\frac{x-1}{2} = \frac{y-3}{2} = \frac{z-5}{-1}$ is

$$5x^2 + 5y^2 + 8z^2 - 8xy + 4xz + 4yz - 6x - 42y - 96z + 225 = 0$$

Sol:- Here radius of cylinder = 3

The eq. of axis of cylinder is

$$\frac{x-1}{2} = \frac{y-3}{2} = \frac{z-5}{-1} \quad \text{--- (1)}$$

Its d.r.s. are 2, 2, -1

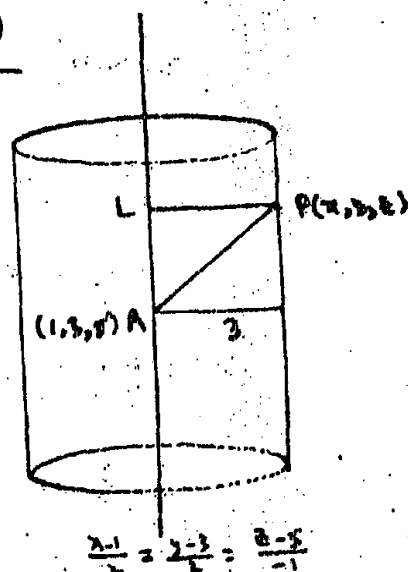
Let $P(x, y, z)$ be any pt. on the surface of cylinder then d.r.s. of AP are $x-1, y-3, z-5$ where A is a pt. on axis & let L be the foot of perp. from P on its axis then
 $|AL|$ = Projection of AP on axis of cylinder

$$|AL| = \frac{2(x-1) + 2(y-3) - 1(z-5)}{\sqrt{(2)^2 + (2)^2 + (-1)^2}}$$

$$= \frac{2x + 2y - z - 2 - 6 + 5}{\sqrt{9}}$$

$$|AL| = \frac{2x + 2y - z - 3}{3}$$

$$|AL|^2 = \frac{1}{9} (2x + 2y - z - 3)^2$$



$$|AL|^2 = \frac{1}{9} \left[(2x+2y) - (z+3) \right]^2$$

$$= \frac{1}{9} \left[(2x+2y)^2 + (z+3)^2 - 2(2x+2y)(z+3) \right]$$

$$= \frac{1}{9} \left[4x^2 + 4y^2 + 8xy + z^2 + 6z + 9 - 2(2xz + 6x + 2yz + 6y) \right]$$

$$= \frac{1}{9} \left[4x^2 + 4y^2 + 8xy + z^2 + 6z + 9 - 4xz - 12x - 4yz - 12y \right]$$

$$|AL|^2 = \frac{1}{9} \left[4x^2 + 4y^2 + z^2 + 8xy - 4yz - 4xz - 12x - 12y + 6z + 9 \right]$$

$$4|AP|^2 = (x-1)^2 + (y-3)^2 + (z-5)^2$$

$$= x^2 - 2x + 1 + y^2 - 6y + 9 + z^2 - 10z + 25$$

$$= x^2 + y^2 + z^2 - 2x - 6y - 10z + 35$$

$$4|LP|^2 = (3)^2 = 9$$

From right angled $\triangle ALP$

$$|AP|^2 = |AL|^2 + |LP|^2$$

$$x^2 + y^2 + z^2 - 2x - 6y - 10z + 35 = \frac{1}{9} (4x^2 + 4y^2 + z^2 + 8xy - 4yz - 4xz - 12x - 12y + 6z + 9)$$

Multiplying both sides by 9

$$9x^2 + 9y^2 + 9z^2 - 18x - 54y - 90z + 315 = 4x^2 + 4y^2 + z^2 + 8xy - 4yz - 4xz - 12x - 12y + 6z + 9$$

$$5x^2 + 5y^2 + 8z^2 - 8xy + 4yz + 4xz - 6x - 42y - 96z + 225 = 0$$

is eq. of gen. right circular cylinder.

Q21 Show that the eq. to the right circular cone with vertex at O , axis OZ & semi vertical angle α is $x^2 + y^2 = z^2 \tan^2 \alpha$.

Sol.

Let $P(x, y, z)$ be any pt. on the Cone then
d.r.s. of OP are $x-0, y-0, z-0$
or x, y, z

Now d.r.s. of OZ -axis (z -axis)
are $0, 0, 1$

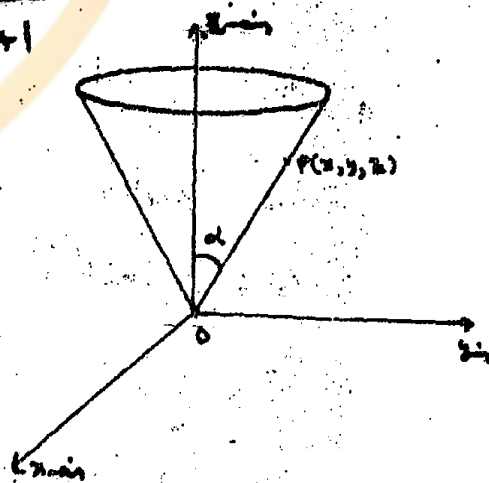
Here α is the semi vertical
angle of Cone then

$$\begin{aligned} \cos \alpha &= \frac{x(0) + y(0) + z(1)}{\sqrt{x^2 + y^2 + z^2} \sqrt{0 + 0 + 1}} \\ &= \frac{0 + 0 + z}{\sqrt{x^2 + y^2 + z^2}} \end{aligned}$$

$$\text{So } \cos \alpha = \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$

$$\Rightarrow \cos^2 \alpha = \frac{z^2}{x^2 + y^2 + z^2}$$

$$\frac{1}{\cos^2 \alpha} = \frac{x^2 + y^2 + z^2}{z^2}$$



$$\sec^2 d = \frac{x^2 + y^2 + z^2}{z^2}$$

$$z^2 \sec^2 d = x^2 + y^2 + z^2$$

$$z^2 (\sec^2 d - 1) = x^2 + y^2$$

$$\boxed{z^2 \tan^2 d = x^2 + y^2}$$

is eq. of req. Cone.

Q22 Show that the general eq. to the Cone of the second degree which passes through the Co-ord. axes is $fyz + gzx + hxy = 0$

Hint

[The general eq. to the Cone of second degree is $ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0$ & this is to be satisfied by the dir. of the Co-ord. axes i.e., by $1, 0, 0$; $0, 1, 0$ & $0, 0, 1$]

Sol.

The general eq. to the Cone of second degree is

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0 \quad \text{--- (A)}$$

Since this Cone passes through

the Co-ord. axes. Therefore the pts. at unit distance from origin of Co-ord. axes lie on Cone (A)

For pt. $(1,0,0)$ Put in (A)

$$a = 0$$

For pt. $(0,1,0)$ Put in (A)

$$b = 0$$

For pt. $(0,0,1)$ Put in (A)

$$c = 0$$

Using these values of a, b & c in (A)

$$0 + 0 + 0 + 2fyz + 2gzx + 2hxy = 0$$

$$\text{or } 2fyz + 2gzx + 2hxy = 0$$

$$\text{or } \boxed{fyz + gzx + hxy = 0} \text{ is eq. of req. Cone.}$$

Q23 Prove that the eq. to the Cone whose vertex is the origin & which passes through the curve of intersection of the plane

$lx + my + nz = p$ & the surface $ax^2 + by^2 + cz^2 = 1$

$$\text{is } ax^2 + by^2 + cz^2 = \left(\frac{lx + my + nz}{p} \right)^2$$

Sol. The eq. of a line through origin is

$$\frac{x-0}{\lambda} = \frac{y-0}{\mu} = \frac{z-0}{\nu}$$

$$\frac{x}{\lambda} = \frac{y}{\mu} = \frac{z}{\nu} = t \quad \text{--- (A)}$$

$$\Rightarrow \left. \begin{aligned} x &= \lambda t \\ y &= \mu t \\ z &= \nu t \end{aligned} \right\}$$

Any pt. on line ① is $(\lambda t, \mu t, \nu t)$

If this pt. lies on the director of req. Cone which is

$$lx + my + nz = p, \quad ax^2 + by^2 + cz^2 = 1$$

Then

$$\left. \begin{aligned} l\lambda t + m\mu t + n\nu t &= p \\ + a\lambda^2 t^2 + b\mu^2 t^2 + c\nu^2 t^2 &= 1 \end{aligned} \right\}$$

$$\text{or } t(l\lambda + m\mu + n\nu) = p \quad \text{--- ①}$$

$$+ t^2(a\lambda^2 + b\mu^2 + c\nu^2) = 1 \quad \text{--- ②}$$

$$\text{from ① } t = \frac{p}{l\lambda + m\mu + n\nu}$$

Put in ②

$$\frac{p^2}{(l\lambda + m\mu + n\nu)^2} (a\lambda^2 + b\mu^2 + c\nu^2) = 1$$

$$p^2(a\lambda^2 + b\mu^2 + c\nu^2) = (l\lambda + m\mu + n\nu)^2$$

Now eliminating λ, μ, ν by using ①

$$p^2(ax^2 + by^2 + cz^2) = (lx + my + nz)^2$$

$$\text{or } ax^2 + by^2 + cz^2 = \left(\frac{lx + my + nz}{p} \right)^2$$

is req. eq. of Cone. _____

Q24 The plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ meets the Co-ord. axes in A, B & C. Prove that an eq. to the Cone generated by lines drawn from the origin to meet the circle ABC is

$$yz\left(\frac{b}{c} + \frac{c}{b}\right) + zx\left(\frac{c}{a} + \frac{a}{c}\right) + xy\left(\frac{a}{b} + \frac{b}{a}\right) = 0$$

Sol First of all we will find the eq. of direction of req. Cone.

For this the Co-ords. of pts. A, B & C are

$$A(a, 0, 0), B(0, b, 0) \text{ \& } C(0, 0, c)$$

We know the eq. of a Sphere through O, A, B & C

$$\text{is } x^2 + y^2 + z^2 - ax - by - cz = 0$$

s. eq. of direction of Cone is

$$x^2 + y^2 + z^2 - ax - by - cz = 0; \quad \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

The eq. of line through origin is

$$\frac{x-0}{\lambda} = \frac{y-0}{\mu} = \frac{z-0}{\nu}$$

$$\text{or } \frac{x}{\lambda} = \frac{y}{\mu} = \frac{z}{\nu} = t$$

$$\Rightarrow \left. \begin{aligned} x &= \lambda t \\ y &= \mu t \\ z &= \nu t \end{aligned} \right\}$$

Any pt. on this line is $(\lambda t, \mu t, \nu t)$

If this pt. lies on direction then

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$$\lambda^2 t^2 + \mu^2 t^2 + \nu^2 t^2 - a\lambda t - b\mu t - c\nu t = 0 \quad \text{--- (1)}$$

$$\frac{\lambda t}{a} + \frac{\mu t}{b} + \frac{\nu t}{c} = 1 \quad \text{--- (2)}$$

$$\text{from (2)} \quad t \left(\frac{\lambda}{a} + \frac{\mu}{b} + \frac{\nu}{c} \right) = 1$$

$$\text{or } t = \frac{1}{\left(\frac{\lambda}{a} + \frac{\mu}{b} + \frac{\nu}{c} \right)}$$

Put in (1)

$$\frac{1}{\left(\frac{\lambda}{a} + \frac{\mu}{b} + \frac{\nu}{c} \right)^2} (\lambda^2 + \mu^2 + \nu^2) - \frac{1}{\left(\frac{\lambda}{a} + \frac{\mu}{b} + \frac{\nu}{c} \right)} (a\lambda + b\mu + c\nu) = 0$$

Multiplying both sides by $\left(\frac{\lambda}{a} + \frac{\mu}{b} + \frac{\nu}{c} \right)^2$

$$\lambda^2 + \mu^2 + \nu^2 - \left(\frac{\lambda}{a} + \frac{\mu}{b} + \frac{\nu}{c} \right) (a\lambda + b\mu + c\nu) = 0$$

Now eliminating λ, μ, ν by using eq. (A)

$$x^2 + y^2 + z^2 - \left(\frac{x}{a} + \frac{y}{b} + \frac{z}{c} \right) (ax + by + cz) = 0$$

$$x^2 + y^2 + z^2 - \left(x^2 + \frac{b}{a}xy + \frac{c}{a}xz + \frac{a}{b}xy + y^2 + \frac{c}{b}yz + \frac{a}{c}xz + \frac{b}{c}yz + z^2 \right) = 0$$

$$x^2 + y^2 + z^2 - \left(x^2 + y^2 + z^2 + xy \left(\frac{a}{b} + \frac{b}{a} \right) + yz \left(\frac{b}{c} + \frac{c}{b} \right) + xz \left(\frac{a}{c} + \frac{c}{a} \right) \right) = 0$$

$$x^2 + y^2 + z^2 - x^2 - y^2 - z^2 - xy \left(\frac{a}{b} + \frac{b}{a} \right) - yz \left(\frac{b}{c} + \frac{c}{b} \right) - xz \left(\frac{a}{c} + \frac{c}{a} \right) = 0$$

Multiplying both sides by -1

$$xy \left(\frac{a}{b} + \frac{b}{a} \right) + yz \left(\frac{b}{c} + \frac{c}{b} \right) + xz \left(\frac{a}{c} + \frac{c}{a} \right) = 0$$

is req. eq. of Cmc.

Q25 Find an eq. to the Cone whose vertex is the origin & directrix, the circle $x=a, y^2+z^2=b^2$.
Show that the trace of the Cone in a plane parallel to xy -plane is a hyperbola.

Sol. Given eq. of directrix is
 $x=a, y^2+z^2=b^2$

The eq. of line through origin is

$$\frac{x-0}{\lambda} = \frac{y-0}{\mu} = \frac{z-0}{\nu}$$

$$\frac{x}{\lambda} = \frac{y}{\mu} = \frac{z}{\nu} = t \quad \text{--- (1)}$$

Any pt. on this line is $(\lambda t, \mu t, \nu t)$

If this pt. lies on dir then

$$\lambda t = a \quad \text{--- (1)}$$

$$\mu^2 t^2 + \nu^2 t^2 = b^2 \quad \text{--- (2)}$$

From (1) $t = \frac{a}{\lambda}$

Put in (2)

$$\frac{a^2}{\lambda^2} \mu^2 + \frac{a^2}{\lambda^2} \nu^2 = b^2$$

$$\frac{a^2}{\lambda^2} (\mu^2 + \nu^2) = b^2$$

$$a^2 (\mu^2 + \nu^2) = b^2 \lambda^2$$

Now eliminating λ, μ, ν by using eq. (1)

$$[a^2 (y^2 + z^2) = b^2 x^2] \text{ is eq. of req. Cone.}$$

For trace of this Cone parallel to xy -plane.

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Put $z = k$ in eq. of Cone.

$$a^2(y^2 + k^2) = b^2x^2$$

$$a^2y^2 + a^2k^2 = b^2x^2$$

$$\text{or } b^2x^2 = a^2y^2 + a^2k^2$$

$$b^2x^2 - a^2y^2 = a^2k^2$$

Dividing both sides by a^2k^2

$$\frac{b^2x^2}{a^2k^2} - \frac{a^2y^2}{a^2k^2} = 1$$

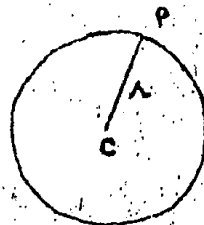
$$\text{or } \frac{x^2}{\frac{a^2k^2}{b^2}} - \frac{y^2}{k^2} = 1$$

$$\text{or } \boxed{\frac{x^2}{\left(\frac{ak}{b}\right)^2} - \frac{y^2}{k^2} = 1} \text{ is eq. trace}$$

which is a hyperbola.

Sphere: It is the locus of a pt. $P(x, y, z)$ which moves in space such that its distance from a fixed pt. remains const.

(i) Fixed pt. is called Centre of sphere.



(ii) Fixed distance $|CP| = r$ is called radius of sphere.

Eq. of a sphere

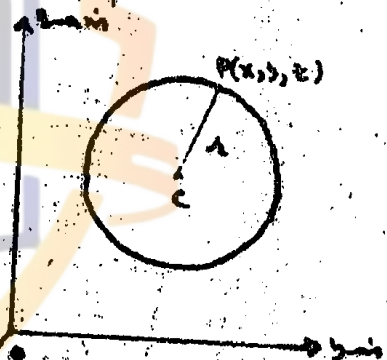
Let $C(a, b, c)$ be the centre of a sphere & r be its radius.

Now take a variable pt.

$P(x, y, z)$ on the sphere

then by def. of sphere

$$|CP| = r$$



$$\sqrt{(x-a)^2 + (y-b)^2 + (z-c)^2} = r$$

or $(x-a)^2 + (y-b)^2 + (z-c)^2 = r^2$ is the eq. of

a sphere with centre (a, b, c) & radius r .

Note If centre is at origin then $(a, b, c) = (0, 0, 0)$

So $x^2 + y^2 + z^2 = r^2$ is eq. of sphere with centre at origin & radius r .

General eq. of a sphere:

The general eq. of a sphere is

$$x^2 + y^2 + z^2 + 2Gx + 2Fy + 2Kz + C = 0 \quad \text{--- ①}$$

Now we find its centre & radius.

Eq. ① can be written as

$$(x^2 + 2Gx) + (y^2 + 2Fy) + (z^2 + 2Kz) = -C$$

Add $G^2 + F^2 + K^2$ on both sides

$$(x^2 + 2Gx + G^2) + (y^2 + 2Fy + F^2) + (z^2 + 2Kz + K^2) = G^2 + F^2 + K^2 - C$$

$$(x+G)^2 + (y+F)^2 + (z+K)^2 = (G^2 + F^2 + K^2 - C)$$

$$\text{or } (x - (-G))^2 + (y - (-F))^2 + (z - (-K))^2 = (\sqrt{G^2 + F^2 + K^2 - C})^2$$

which is the eq. of a sphere.

Its centre is $(-G, -F, -K)$

Its radius is $\sqrt{G^2 + F^2 + K^2 - C}$

Note An eq. of the form

$$ax^2 + by^2 + cz^2 + 2Gx + 2Fy + 2Kz + C = 0$$

will be a sphere if

- (i) $a = b = c$
- (ii) $a, b \text{ \& } c$ are of same signs.